

Computer vision and assistive navigation

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Software engineer with research experience at the University of Algarve's Vision Laboratory. My academic work focused on computer vision to support the mobility of blind and visually impaired people, from real-time path and obstacle detection to indoor navigation.

Research experience

Computer Vision Researcher and Research Fellow

Vision Laboratory, University of Algarve

2008 - 2013

Faro, Portugal. Developed human-vision models and active-vision algorithms in C/C++ and Matlab. Research member of ISR-Lisbon/IST, collaborating with the University of Trás-os-Montes and Alto Douro (UTAD).

SmartVision. Contributed to a portable navigation aid combining computer vision, geographic information and localisation. Detected walkable paths and static or moving obstacles beyond the reach of the white cane.

Blavigator. Worked on path following, obstacle detection and indoor landmarks, including lateral-door detection and sound feedback for navigation guidance.

Research grants and awards

SmartVision research grant

Active vision for the blind.

PTDC/EIA/73633/2006

Blavigator research grant

A cheap and reliable navigation aid for the blind.

RIPD/ADA/109690/2009

Doctoral research grant

Robust visual recognition of objects and environments for blind persons.

SFRH/BD/82310/2011

CERATONIA award grant

The Amateur Painter.

3.70.13.001

Publications

Journal articles

2015 Computer vision and GIS for the navigation of blind persons in buildings ↗

M. Serrão, S. Shahrabadi, M. Moreno, **J. T. José**, J. I. Rodrigues, J. M. F. Rodrigues & J. M. H. du Buf

Universal Access in the Information Society, 14, 67-80. March 2015; online 19 February 2014. DOI: [10.1007/s10209-013-0338-8](https://doi.org/10.1007/s10209-013-0338-8)

2011 The SmartVision local navigation aid for blind and visually impaired persons ↗

J. José, M. Farrajota, J. M. F. Rodrigues & J. M. H. du Buf

International Journal of Digital Content Technology and its Applications, 5(5), 362-375. DOI: [10.4156/jdcta.vol5.issue5.40](https://doi.org/10.4156/jdcta.vol5.issue5.40)

2011 The SmartVision Navigation Prototype for Blind Users ↗

J. M. H. du Buf, J. Barroso, J. M. F. Rodrigues, H. Paredes, M. Farrajota, H. Fernandes, **J. José**, V. Teixeira & M. Saleiro

International Journal of Digital Content Technology and its Applications, 5(5), 351-361. DOI: [10.4156/jdcta.vol5.issue5.39](https://doi.org/10.4156/jdcta.vol5.issue5.39)

Conference papers

2012 Realtime Local Navigation for the Blind: Detection of Lateral Doors and Sound Interface ↗

M. Moreno, S. Shahrabadi, **J. José**, J. M. H. du Buf & J. M. F. Rodrigues

DSAI 2012. Procedia Computer Science, 14, 74-82. DOI: [10.1016/j.procs.2012.10.009](https://doi.org/10.1016/j.procs.2012.10.009)

2012 Visual Navigation for the Blind - Path and Obstacle Detection ↗

J. José, J. M. H. du Buf & J. M. F. Rodrigues

1st International Conference on Pattern Recognition Applications and Methods (ICPRAM), 515-519. DOI: [10.5220/0003711405150519](https://doi.org/10.5220/0003711405150519)

2010 A vision system for detecting paths and moving obstacles for the blind ↗

J. José, M. Farrajota, J. M. F. Rodrigues & J. M. H. du Buf

3rd International Conference on Software Development for Enhancing Accessibility and Fighting Info-exclusion (DSAI), Oxford, 175-182.

2010 The SmartVision Navigation Prototype for the Blind ↗

J. M. H. du Buf, J. Barroso, J. M. F. Rodrigues, H. Paredes, M. Farrajota, H. Fernandes, **J. José**, V. Teixeira & M. Saleiro

3rd International Conference on Software Development for Enhancing Accessibility and Fighting Info-exclusion (DSAI), Oxford, 167-174.

Conference abstract

2012 Blavigator: a navigation aid for blind persons ↗

J. Jose, M. Moreno, J. Pinilla-Dutoit, J. Rodrigues & J. Du Buf

ECVP 2012. Perception, 41(1_suppl), p. 99, abstract 97. Supplement DOI: [10.1177/03010066120410s101](https://doi.org/10.1177/03010066120410s101)

Book

2013 Real-time path and obstacle detection for blind persons ↗

João Tiago Pereira Nunes José

LAP LAMBERT Academic Publishing. ISBN 978-3-659-45730-2.

Education and teaching

Education

Doctoral studies in Computer Science Engineering

University of Algarve

Sep 2011 - Oct 2013

Completed the programme expected for the first two years; doctorate not completed and studies subsequently suspended. Research covered robust visual recognition of objects and environments for blind users, literature review and algorithm comparison, an initial object-recognition model, and hand and arm segmentation.

MSc in Computer Science Engineering

University of Algarve

Sep 2008 - Jun 2011

Degree grade: 16/20. Dissertation grade: 18/20.

Dissertation (2010): *Real-time path and obstacle detection for blind persons* ↗

Real-time path and obstacle detection on a small netbook using C/C++, OpenCV and Qt. Supervisor: Prof. Johannes Martinus Hubertina du Buf.

BSc in Computer Science Engineering

University of Algarve

Sep 2003 - Jul 2008

Selected project: a web content-management system for football news and results, using HTML, PHP, JavaScript, PostgreSQL and PL/pgSQL. Project grade: 19/20.

Studies in Electronics Engineering

University of Algarve

Sep 2001 - Jul 2003

Changed major to Computer Science Engineering; no degree awarded in Electronics Engineering.

Teaching

Cisco CCNA Exploration Instructor

Cisco Academy, FCT - University of Algarve

Nov 2008 - Jun 2009

Faro, Portugal. Taught the four CCNA Exploration modules:

Network Fundamentals. Network models and protocols, Ethernet, IP addressing and subnetting.

Routing Protocols and Concepts. Routing fundamentals, VLSM, RIP, OSPF and EIGRP.

LAN Switching and Wireless. Switch configuration, VLANs, trunking, spanning tree and WLANs.

Accessing the WAN. WAN technologies, ACLs, PPP/PPPoE, Frame Relay and VPN concepts.

Training and qualifications

Certificate of Pedagogical Competences (CCP) · 2015. IEFEP, March 2015. Initial trainer training in pedagogical methods and competences at Training House, Faro, January-March 2015.

Cisco Certified Network Associate (CCNA) · 2008. Cisco Academy, FCT - University of Algarve, October 2007-May 2008. Modules 1-4; average grade: 90%.

Professional Scrum Master I · 2018. Scrum.org, March 2018.

Professional experience

Independent Software Engineer

JTJ Software · Netherlands and Portugal

Nov 2022 - Present

LVNL SIM Builder · Jun 2024 - Feb 2026. Contributed across the full stack of a web-based ATM scenario platform built on an adapted BlueSky simulator, combining development and Scrum Master responsibilities.

LVNL iCAS SIM · Nov 2022 - Dec 2025. Developed and maintained the simulation and training environment for the Netherlands' next-generation air traffic management system, including integration with peripheral systems.

Portuguese real-estate software · Jan 2026 - Present. Product discovery, architecture and implementation for the Portuguese real-estate market.

C++ Software Developer

Accent Pointe B.V. · Netherlands

May 2021 - Nov 2022

Developed trading software integrated with brokers and market-data feeds; modernised a legacy MFC and Microsoft RPC codebase.

C++ Software Developer

ABN AMRO Clearing Bank · Netherlands

Nov 2016 - May 2021

Delivered financial risk-management software and integrated service interfaces, messaging and authentication components.

Software Engineer

Frinch · Faro, Portugal / Remote

Apr 2016 - Oct 2016

Developed Product Information Management software with a distributed team working with the company's Munich headquarters.

Software Engineer and Lead Researcher

Evision Industry Software · Faro, Portugal

Oct 2013 - Sep 2015

Developed algorithms for text detection, raster-to-vector conversion and automatic interpretation of piping and instrumentation diagrams and AutoCAD files, including pipe and instrument recognition. Worked with the team in The Hague using C# and TypeScript.

Invited Expert

SEF · Faro, Portugal

May 2007 - Aug 2007

Participated in the quality assessment of the electronic border-control system installed at Faro Airport.

Technical skills

Research and computer vision. C/C++, Matlab, OpenCV and Qt; real-time path and obstacle detection, visual object recognition, text detection and raster-to-vector conversion.

Software engineering. C++, Python, C, Go, Ada, Bash, C# and TypeScript; HTML, PHP and JavaScript; REST APIs, ZeroMQ, Kafka and Protobuf; PostgreSQL, PL/pgSQL and Ingres; Git, CMake, GCC and GDB; pytest and C++ testing tools; Linux, Docker and OpenShift.

Languages

Portuguese · Native English · Fluent Dutch · Intermediate Spanish · Good